Volume 8 – Machine Learning & Predictive Intelligence Design

AI Quant Trading Platform

Version: 1.0

Document Type: Machine Learning Architecture

Classification: Advanced Intelligence Layer

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1. Purpose

The Machine Learning (ML) subsystem enhances the platform by discovering patterns in historical market behavior and strategy outcomes. Unlike the rule-based AI Decision Engine, the ML subsystem focuses on prediction, evaluation, and continuous improvement using validated historical data.

The ML subsystem shall **support** trading decisions. It shall not bypass the Strategy Engine or Risk Engine.

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2. Objectives

The ML subsystem shall:

- Learn from historical trade outcomes
- Rank trading opportunities
- Predict probability of success
- Classify market regimes
- Detect anomalies
- Improve recommendation quality
- Monitor model performance
- Support research and experimentation

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3. ML Architecture

```
Market Data
        |
Historical Database
        |
Feature Engineering
```

```
        |
Feature Store
        |
Training Pipeline
        |
Model Registry
        |
Validation Pipeline
        |
Inference Service
        |
Prediction API
        |
AI Decision Engine
```

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4. Data Sources

Market Data

- OHLCV
- Tick Data
- Order Book
- Volume
- Spread
- Funding Rate
- Open Interest
- Liquidation Events

Technical Indicators

- EMA
- SMA
- RSI
- MACD
- ATR
- ADX
- VWAP
- SuperTrend
- Bollinger Bands
- Volume Profile
- MFI
- OBV

Trade History

- Entry Price

- Exit Price
- Stop Loss
- Take Profit
- Holding Time
- Position Size
- Strategy Used
- Trade Outcome

Portfolio Data

- Balance
- Drawdown
- Exposure
- Allocation

Context Data

- BTC Trend
- ETH Trend
- Stablecoin Dominance
- Session Timing
- Market Regime
- Exchange Status

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5. Feature Engineering

Features include:

Price Features

- Returns
- Volatility
- Momentum
- Trend Strength

Indicator Features

- RSI
- EMA Distance
- MACD Histogram
- ATR
- ADX

Market Features

- Liquidity
- Spread

- Volume Spike
- Order Book Imbalance

Strategy Features

- Active Strategy
- Timeframe
- Risk Profile

Portfolio Features

- Exposure
- Current Drawdown
- Open Positions

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6. Feature Store

The Feature Store maintains standardized inputs for all models.

Requirements:

- Versioned features
- Timestamped records
- Reusable across models
- Validation before training

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7. Model Categories

Classification Models

Purpose:

Predict whether a setup satisfies predefined quality criteria.

Regression Models

Purpose:

Estimate expected movement or outcome-related metrics.

Ranking Models

Purpose:

Rank candidate opportunities.

Anomaly Detection

Purpose:

Detect abnormal market behavior.

Market Regime Models

Purpose:

Classify market state:

- Bull
- Bear
- Range
- High Volatility
- Low Volatility

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8. Candidate Algorithms

Traditional

- Logistic Regression
- Random Forest
- Gradient Boosting
- XGBoost
- LightGBM

Deep Learning

- LSTM
- Transformer-based Time Series
- Temporal Convolutional Networks (future)

Experimental

- Reinforcement Learning (research only)
- Ensemble Models

Algorithm selection shall be evidence-based using validation metrics.

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9. Model Training Pipeline

Data Collection

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Cleaning

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Feature Engineering

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Training

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Validation

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Performance Evaluation

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Approval

↓

Deployment

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Monitoring

Only validated models may enter production.

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10. Training Schedule

Options:

- Manual
- Weekly
- Monthly
- Triggered by sufficient new data

Production retraining requires validation and approval.

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11. Validation

Metrics

Classification

- Precision
- Recall
- F1 Score
- ROC-AUC

Regression

- MAE
- RMSE

Ranking

- NDCG
- MAP

Operational

- Inference Latency
- Stability
- Drift

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12. Model Registry

Every deployed model stores:

- Model ID
- Version
- Training Date
- Dataset Version
- Hyperparameters
- Performance Metrics
- Approval Status
- Rollback Version

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13. Inference Service

Responsibilities

- Load active model
- Validate input features

- Produce prediction
- Return confidence
- Record inference logs

The service must remain stateless for horizontal scaling.

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14. Drift Detection

Monitor:

- Feature Drift
- Data Drift
- Prediction Drift
- Performance Drift

When thresholds are exceeded:

- Alert administrators
- Flag model for review
- Prevent automatic promotion

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15. Explainability

Each prediction should provide supporting factors where feasible.

Example:

- Strong higher-timeframe trend
- Elevated volume
- Low spread
- Momentum confirmation

Explainability improves transparency and user trust.

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16. Online Learning

Direct online modification of production models is disabled by default.

Instead:

- Collect feedback
- Store new observations
- Evaluate offline
- Retrain under controlled conditions
- Promote only after validation

17. Reinforcement Learning (Research)

Future research module.

Potential uses:

- Position management
- Exit optimization
- Strategy parameter tuning

Constraints:

- Isolated research environment
- Never directly controls live trading
- Requires extensive validation

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18. Security

- Model artifacts are signed
- Feature validation before inference
- Dataset access controlled
- Audit logs for model deployment
- Version rollback supported

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19. Monitoring

Monitor:

- Prediction latency
- Model accuracy trends
- Drift indicators
- Error rates
- Feature availability
- Resource usage

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20. Failure Handling

If the ML subsystem is unavailable:

- Continue operating using the rule-based AI Decision Engine
- Record incident
- Notify administrators

• Prevent incomplete predictions from influencing decisions

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21. Design Principles

• Models assist; they do not override risk controls.
• Production models require validation before deployment.
• Every prediction is traceable to a model version.
• Retraining is controlled and auditable.
• System behavior remains deterministic when ML is unavailable.
• Safety has higher priority than prediction accuracy.